University of Ottawa

Department of Mathematics and Statistics

MAT 2143B : Introduction to Group Theory

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Final Exam

April 20, 2022

Surname _____ First Name _____

Student # _____ DGD _____

Instructions:

(a) You have 180 minutes to complete this exam and submit your PDF file on BrightSpace.
(b) All work to be considered for grading should be written on your solution sheets. You should clearly indicate where the solution of each question is written.
(c) No notes, books, scrap paper, calculators or other electronic devices are allowed.
(d) You are strongly recommended to write in **pen**, not pencil.

Please do not write in the table below.

Question	1	2	3	4	5	6	7	8	9	10	Total
Maximum	5	3	3	5	6	4	6	4	6	2	44
Grade											

1. (a) **[1 point]** Write down the definition of a prime number.

Solution: An integer p is called prime if $p > 1$ and the only divisors of p are ± 1 and $\pm p$.

(b) **[2 points]** Can $16^n - n^2$ be a prime number for any integer $n > 1$? Justify your answer.

Solution: We have $16^n - n^2 = 2^{4n} - n^2 = (2^{2n} - n)(2^{2n} + n)$. The two factors are both bigger than 1. Hence $16^n - 1$ is a composite number.

(c) **[2 points]** An integer x is called a *perfect cube* if $x = a^3$ for some $a \in \mathbb{Z}$. Let x and y be co-prime integers such that xy is a perfect cube. Prove that both x and y are perfect cubes.

Solution: The prime factorization of xy is $p_1^{3k_1} \cdots p_r^{3k_r}$ for primes $p_1, \ldots, p_r$ and non-negative integers $k_1, \ldots, k_r$. It follows that the prime factorizations of x and y are:

$$x = p_1^{a_1} \cdots p_r^{a_r} \quad , \quad y = p_1^{b_1} \cdots p_r^{b_r},$$

where $a_i + b_i = 3k_i$ for $1 \leq i \leq r$. Since x and y are coprime, there is no prime that divides both x and y. It follows that for every i we have either $a_i = 0$ (hence $b_i = 3k_i$), or $b_i = 0$ (hence $a_i = 3k_i$). Consequently, both x and y are perfect cubes.

2. Consider the binary operation $\star$ on $\mathbb{N} := \{1, 2, 3, \ldots\}$, defined as follows:

$$a \star b = \max\{a, b\}.$$

(a) [**2 points**] Prove that $1 \in \mathbb{N}$ is the unique identity element for $\mathbb{N}$ with respect to $\star$.

Solution: For every $a \in \mathbb{N}$, we have $1 \star a = \max\{1, a\} = a$ since $a \geq 1$. Similarly, $a \star 1 = a$. Thus 1 is an identity element.

For uniqueness, note that if e is an identity element, then $e = 1 \star e = 1$.

(b) [**1 point**] Which elements of $\mathbb{N}$ are invertible with respect to the binary operation $\star$? You should justify your answer.

Solution: Suppose $a \in \mathbb{N}$ is invertible. Then $a \star a' = 1$ for some $a' \in \mathbb{N}$. Thus $\max\{a, a'\} = 1$. Since $a, a' \geq 1$, it follows that $a = a' = 1$.

3. [**3 points**] Recall that $U(n)$ is the group of units of $\mathbb{Z}_n$. For which integers $n \geq 2$ is it true that $n+1 \in U(2n)$? You should justify your answer.

Solution: Suppose that $d|2n$ and $d|(n+1)$. Then $d|(2(n+1)-2n)$, hence $d|2$. Now if n is even then $n+1$ is odd and from $d|2$ and $d|n+1$ it follows that $d=1$, hence $\gcd(n+1, 2n) = 1$. Thus in this case $n+1 \in U(2n)$. However, if n is odd then $n+1$ is even, and it is clear that $n+1$ and $2n$ are not coprime. Therefore $n+1 \notin U(2n)$.

4. (a) **[1 point]** Write down the definition of a normal subgroup of a group G.

Solution: A subgroup H of G is called normal if $gH = Hg$ for every $g \in G$.

(b) **[2 points]** Let H be the cyclic subgroup of the permutation group S_8 that is generated by the permutation

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 3 & 4 & 5 & 6 & 7 & 8 & 1 & 2 \end{pmatrix}.$$

How many elements does H have?

Solution: It is not difficult to see that $\sigma = (1\ 3\ 5\ 7)(2\ 4\ 6\ 8)$. Thus

$$\sigma^2 = (1\ 5)(3\ 7)(2\ 6)(4\ 8)\ ,\ \sigma^3 = (1\ 7\ 5\ 3)(2\ 8\ 6\ 4),$$

and $\sigma^4 = \varepsilon$. It follows that $|H| = 4$.

(c) **[1 point]** Is H a normal subgroup of S_8? You should justify your answer.

Solution: H is not a normal subgroup. For example for $\tau = (1\ 2)$ we have

$$\tau\sigma\tau^{-1} = (1\ 4\ 6\ 8)(2\ 3\ 5\ 7),$$

which is not among the 4 elements of H that were listed above. Thus $\tau H \tau^{-1} \neq H$.

(d) **[1 point]** The subgroup H has another generator τ. Write down τ in the array form:

$$\tau = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ _ & _ & _ & _ & _ & _ & _ & _ \end{pmatrix}$$

Solution: H is a cyclic group of order 4, hence it is isomorphic to $\mathbb{Z}_4$. But $\mathbb{Z}_4$ has two generators: 1 and 3. We can assume σ corrsponds to 1, so that σ^3 corresponds to 3. Thus the other generator of H is σ^3, which is given by:

$$\tau = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 7 & 8 & 1 & 2 & 3 & 4 & 5 & 6 \end{pmatrix}$$

5. (a) **[3 points]** Compute the order of $(2,3) \in \mathbb{Z}_{28} \times \mathbb{Z}_{30}$. You should justify your answer.

Solution: Suppose that the order is k. Thus $(2k, 3k) = (0,0)$, hence $28|2k$ and $30|3k$. It follows that

$$14|k \text{ and } 10|k.$$

The smallest positive integer divisible by both 14 and 10 is 70. It follows that $k \geq 70$. It is easily seen that $70(2,3) = (0,0)$, hence the answer is 70.

(b) **[2 points]** Let G be a group with an odd number of elements. Prove that for every $g \in G$ we have $o_G(g) = o_G(g^2)$.

Hint. First write a formula for $o_G(g^2)$ in terms of $o_G(g)$.

Solution: We have $o(g^2) = o(g)/\gcd(2, o(g))$. Since $|G|$ is odd and $o(g)$ is a divisor of $|G|$ (corollary to Lagrange's theorem), it follows that $\gcd(2, o(g)) = 1$. Hence $o(g^2) = o(g)$.

(c) **[1 point]** Is the assertion of part (b) true for a group with an even number of elements? Justify your answer.

Solution: No. For exmaple in $\mathbb{Z}_2 = \{0, 1\}$ we have $o(1) = 2$ but $o(1+1) = 1$ since $1 + 1 = 0$ is the identity element of $\mathbb{Z}_2$.

6. [**4 points**] For each of the following statements, write **T** if the statement is true, and write **F** if the statement is false. You will receive .5 points for a correct answer, and -.25 points for an incorrect answer. (If you leave the answer space blank, you will not receive any negative points.) You do not need to justify your answers. **For each answer on your answer sheet, you should clearly indicate the part that you are answering**.

(a)____ Every $2d$-cycle, where $d \in \mathbb{N}$, is an even permutation.

(b)____ If H and K are two subgroups of a group G then we always have $HK = KH$.

(c)____ The permutation group S_{20} has a subgroup that is isomorphic to the dihedral group D_{10}.

(d)____ A Cartesian product of finitely many cyclic groups is always a finitely generated abelian group.

(e)____ In public-key cryptography, the parties share a secret decryption procedure.

(f)____ A permuation in S_n is uniquely determined by its cycle decomposition.

(g)____ Every row of the Cayley table of a group G contains every element of G exactly once.

(h)____ The intersection of two normal subgroups is always a normal subgroup.

Solution: F, F, T, T, F, T, T, T

7. (a) **[3 points]** Describe all of the subgroups of $\mathbb{Z}_{20}$ explicitly. How many subgroups are there in total?

Solution: The subgroups of Z_{20} are of the form $d\mathbb{Z}_{20}$ where $d \geq 1$ and $d|20$. Thus $d = 1, 2, 4, 5, 10, 20$. We obtain

$$d = 1 \Rightarrow \mathbb{Z}_{20} = \{0, 1, \ldots, 19\}$$
$$d = 2 \Rightarrow 2\mathbb{Z}_{20} = \{0, 2, 4, 6, 8, 10, 12, 14, 16, 18\}$$
$$d = 4 \Rightarrow 4\mathbb{Z}_{20} = \{0, 4, 8, 12, 16\}$$
$$d = 5 \Rightarrow 5\mathbb{Z}_{20} = \{0, 5, 10, 15\}$$
$$d = 10 \Rightarrow 10\mathbb{Z}_{20} = \{0, 10\}$$
$$d = 2 \Rightarrow 20\mathbb{Z}_{20} = \{0\}$$

Thus there are 6 subgroups.

(b) **[3 points]** Let G be a group and let H be a subgroup of G. Prove that for two cosets aH and bH, we have $aH = bH$ if and only if $a^{-1}b \in H$.

Solution: See proof of Theorem 9.3 in the lecture notes.

8. (a) **[1 point]** Write down the statement of Lagrange's Theorem.

Solution: Let G be a finite group, and let H be a subgroup of G. Then

- $|H|$ divides $|G|$.
- The number of distinct right cosets of H in G is equal to $\frac{|G|}{|H|}$.
- The number of distinct left cosets of H in G is equal to $\frac{|G|}{|H|}$.

(b) **[3 points]** Consider $\mathbb{Q}$, the group of rational numbers (with addition as the binary operation). Let G be the subset of $\mathbb{Q}$ defined as follows:

$$G = \left\{ \frac{k}{2^m 5^n} \ : \ k \in \mathbb{Z} \text{ and } m, n \in \mathbb{N} \right\}.$$

Using the subgroup test prove that G is a subgroup of $\mathbb{Q}$.

Solution: Clearly G is nonempty.

First note that the inverse of $x \in \mathbb{Q}$ is $-x$. Now choose two elements $a = \frac{k}{2^m 5^n}$ and $b = \frac{k'}{2^{m'} 5^{k'}}$ in G. It suffices to verify that $a - b \in G$. This can be proved as follows:

$$a - b = \frac{k}{2^m 5^n} - \frac{k'}{2^{m'} 5^{n'}} = \frac{2^{m'} 5^{n'} k - 2^m 5^n k'}{2^{m+m'} 5^{n+n'}} = \frac{k_1}{2^{m_1} 5^{n_1}},$$

where $k_1 = 2^{m'} 5^{n'} k - 2^m 5^n k' \in \mathbb{Z}$, and $m_1 = m + m'$ and $n_1 = n + n'$ are positive integers. It follows that $a - b \in G$.

9. (a) **[1 point]** Write down the definition of the kernel of a group homomorphism. **Solution:** For a homomorphism $\phi : G \to H$ we have $\ker \phi = \{g \in G \phi(g) = e_H\}$

(b) **[2 points]** Prove that the kernel of a group homomorphism $\phi : G \to H$ is always a normal subgroup of G.

Solution: For $g, h \in \ker \phi$ we have $\phi(gh^{-1}) = \phi(g)\phi(h)^{-1} = e_H e_H^{-1} = e_H$, hence $gh^{-1} \in \ker \phi$. Thus by the Subgroup Test it follows that $\ker \phi$ is a subgroup. For normality, for $g \in \ker \phi$ and $x \in G$ we have

$$\phi(xgx^{-1}) = \phi(x)\phi(g)\phi(x)^{-1} = \phi(x)e_H\phi(x)^{-1} = \phi(x)\phi(x)^{-1} = e_H,$$

hence $xgx^{-1} \in \ker \phi$. It follows that $\ker \phi$ is normal.

(c) **[2 points]** Compute the kernel of the group homomorphism $\phi : U(15) \to U(15)$, given by $\phi(x) = x^2$, explicitly.

Solution: We have

$$U(15) = \{1, 2, 4, 7, 8, 11, 13, 14\}.$$

Now we have $\phi(1) = 1^2 = 1$.
$\phi(2) = 2^2 = 4$.
$\phi(4) = 4^2 = 16 = 15 \times 1 + 1 = 1$.
$\phi(7) = 7^2 = 49 = 15 \times 3 + 4 = 4$.
$\phi(8) = 8^2 = (15 - 7)^2 = 15(15 - 2 \times 7) + 7^2 = 4$
and similarly,
$\phi(11) = 11^2 = (15 - 4)^2 = 4^2 = 1$.
$\phi(13) = 13^2 = (15 - 2)^2 = 2^2 = 4$.
$\phi(14) = 14^2 = (15 - 1)^2 = 1^2 = 1$.
Thus $\ker(\phi) = \{1, 4, 11, 14\}$

(d) **[1 point]** Set $K = \ker \phi$ where ϕ is the homomorphism of part (c). The quotient G/K is isomorphic to a familiar group. Which group is it? You should justify your answer.

Solution: The quotient has 2 elements, hence it is isomorphic to $\mathbb{Z}_2$.

10. [**2 points**] Determine all of the group homomorphisms $\phi : \mathbb{Z}_6 \to \mathbb{Z}_2 \times \mathbb{Z}$. You should justify your answer.

Solution: If $\phi(1) = (a, b)$, then (a, b) should be an element of finite order (because so is $1 \in \mathbb{Z}_6$). Hence we should have $b = 0$. Now a can be either 0 or 1 in $\mathbb{Z}_2$, and both choices yield a group homomorphism. Thus there are precisely two such homomorphisms:

$$\phi(a) = (a \bmod 2, 0)$$

and

$$\phi(a) = (0, 0).$$